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Task 3 Lab Report:

Lab 3: Uninformed Search Algorithms

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Course: Artificial Intelligence Lab

1. Objective

To understand and implement five uninformed search algorithms: BFS, DFS, DLS, IDDFS and UCS.

2. BFS (Breadth-First Search)

BFS explores nodes level by level. It uses a Queue (FIFO).

- Starting Node: A
- Goal Node: G
- Visit Order: A, B, C, D, E, F, G
- Final Path: A → B → E → G

3. DFS (Depth-First Search)

DFS explores one branch deeply before moving to another branch. It uses a Stack (LIFO).

- Starting Node: A
- Goal Node: G
- Visit Order: A, B, D, E, G
- Final Path: A → B → E → G

4. DLS (Depth-Limited Search)

DLS is a depth-first search with a fixed depth limit.

6. UCS (Uniform Cost Search)

UCS selects the path with the lowest total cost using a Priority Queue.

- Final Path: A → B → E → G
- Total Cost: 3 (using the example weights)

7. Algorithm Comparison

Algorithm	Data Structure
BFS	Queue
DFS	Stack
DLS	Stack
IDDFS	Repeated DLS
UCS	Priority Queue

8. Conclusion

In this lab, I learned how BFS, DFS, DLS, IDDFS and UCS work. BFS explores level by level, DFS explores deeply, DLS uses a depth limit, IDDFS increases the depth limit, and UCS finds the minimum-cost path. These algorithms are useful for solving search problems in Artificial Intelligence.